MIL-C-38240/24B(USAF)
9 June 1969
Superseding
MIL-C-38240/24A(USAF)
9 May 1968

MILITARY SPECIFICATION

COMPUTER, ALTITUDE, ALTITUDE ENCODING CPU-66/A-24, T-2B

1. SCOPE

1.1 This specification covers the requirements for an altitude encoding, altitude transducer computer, designated CPU-66/A-24.

*2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-C-38240 Computer, Altitude, Altitude Encoding CPU-66, General Specification for and MTU-39/A Mounting Tray

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 General specification. The requirements of MIL-C-38240 apply as requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:09Z
Check the source to verify that this is the current version before use.

4.2 Classification of tests. The inspection and testing of computers shall be in accordance with MIL-C-38240 with the exceptions and additions specified herein.

4.3 Test methods

4.3.1 Scale error. The test points and scale error tolerances shall be as specified in table I. Scale error tests shall be conducted in both ascending and descending directions. The static pressure error corrections shall be in accordance with table II. The static pressure error curve is shown on figure I to further clarify the configuration.

4.3.1.1 Critical areas. Critical areas are at all altitudes up to and including 50,000 feet and at all speeds up to and including 0.75 mach.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-38240.

6. NOTES

6.1 Intended use. The CPU-66/A-24 computer covered by this specification is intended for use in T-2B aircraft.

6.2 Ordering data. Ordering data will be in accordance with MIL-C-38240.

6.3 Marginal indicia. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6610-F096

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:09Z
Check the source to verify that this is the currentversion before use.

TABLE I. Test Points and Scale Error Tolerances

CALIBRATED PRESSURE ALTITUDE	INDICATED MACH NO.	INDICATED STATIC PRESSURE (P_{S_I}) IN. HG	TOTAL PRESSURE (P_T) IN. HG	DIFFERENTIAL PRESSURE (P_T-P_{S_I}) IN. HG	SCALE ERROR TOLERANCE					
					ROOM TEMP		-35°C, 71°C		-54°C	
					± FT	SYN ± DEG	± FT	SYN ± DEG	± FT	SYN ± DEG
* 0	0	29.9213	29.9213	0	26	0.95	41	1.48	41	1.48
0	0.15	29.9602	30.4346	0.4744	29	1.05	44	1.58	44	1.58
0	0.25	30.0020	31.3353	1.3333	28	1.02	43	1.55	43	1.55
0	0.40	30.0948	33.6023	3.5076	37	1.32	52	1.87	52	1.87
** 0	0.50	30.2534	35.8868	5.6335	43	1.56	58	2.09	58	2.09
* 0	0.70	31.0194	43.0269	12.0075	87	3.12	102	3.67	102	3.67
1,000	0	28.8557	28.8557	0	26	0.95	43	1.55	43	1.55
2,000	0	27.8210	27.8210	0	26	0.95	45	1.62	46	1.66
3,000	0	26.8167	26.8167	0	26	0.95	48	1.73	49	1.76
* 5,000	0.7	25.8095	35.8004	9.9908	85	3.04	113	4.07	115	4.14
10,000	0.4	20.6962	23.1084	2.4121	36	1.29	76	2.73	81	2.92
* 10,000	0.6	20.9923	26.7760	5.7835	56	2.03	96	3.43	101	3.64
10,000	0.7	21.3321	29.5897	8.2576	82	2.97	122	4.39	127	4.57
** 10,000	0.75	21.5934	31.3595	9.7661	90	3.25	130	4.68	135	4.86
20,000	0.3	13.7941	14.6828	0.8887	53	1.91	93	3.35	103	3.71
20,000	0.4	13.8298	15.4417	1.6119	60	2.16	100	3.60	110	3.96
20,000	0.5	13.9027	16.4915	2.5888	66	2.38	106	3.82	116	4.18
20,000	0.65	14.1227	18.7594	4.6367	89	3.22	129	4.64	139	5.00
20,000	0.75	14.4293	20.9553	6.5260	111	3.98	151	5.44	161	5.80
30,000	0.35	8.9209	9.7096	0.7887	81	2.91	121	4.36	136	4.90
30,000	0.45	8.9583	10.2937	1.3354	87	3.12	127	4.58	142	5.11
30,000	0.55	9.0178	11.0761	2.0583	95	3.42	135	4.86	150	5.40
** 30,000	0.65	9.1262	12.1225	2.9963	111	4.00	151	5.44	166	5.98
* 30,000	0.75	9.3243	13.5415	4.2172	131	4.71	171	6.16	186	6.70
* 40,000	0.4	5.5701	6.2193	0.6492	109	3.92	149	5.36	169	6.08
40,000	0.5	5.5895	6.6421	1.0427	114	4.10	154	5.54	174	6.26
40,000	0.6	5.6499	7.2064	1.5565	125	4.52	165	5.94	185	6.66
40,000	0.7	5.7412	7.9636	2.2224	146	5.27	186	6.70	206	7.41
* 50,000	0.45	3.4527	3.9674	0.5147	136	4.90	176	6.33	201	7.23
** 50,000	0.6	3.4938	4.4564	0.9626	150	5.42	190	6.84	215	7.74
* 50,000	0.75	3.5938	5.2192	1.6254	178	6.40	218	7.85	243	8.75

* Test points at which low temperature operation, high temperature operation, low temperature exposure and high temperature exposure shall be conducted.

** Test points at which vibration error, voltage variation, vibration failure, temperature-altitude, humidity, salt fog, acceleration, dust, mechanical shock and pressurization tests shall be conducted.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T12:09Z
Check the source to verify that this is the current version before use.

TABLE II. Static Position Error Correction

M_I	P_{S_I}/P_S
0.00	1.0000
0.10	1.0005
0.15	1.0013
0.20	1.0020
0.25	1.0027
0.30	1.0032
0.35	1.0040
0.40	1.0058
0.45	1.0082
0.50	1.0111
0.55	1.0149
0.60	1.0202
0.65	1.0271
0.70	1.0367
0.75	1.0494
0.95	1.0500

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:09Z
Check the source to verify that this is the current version before use.

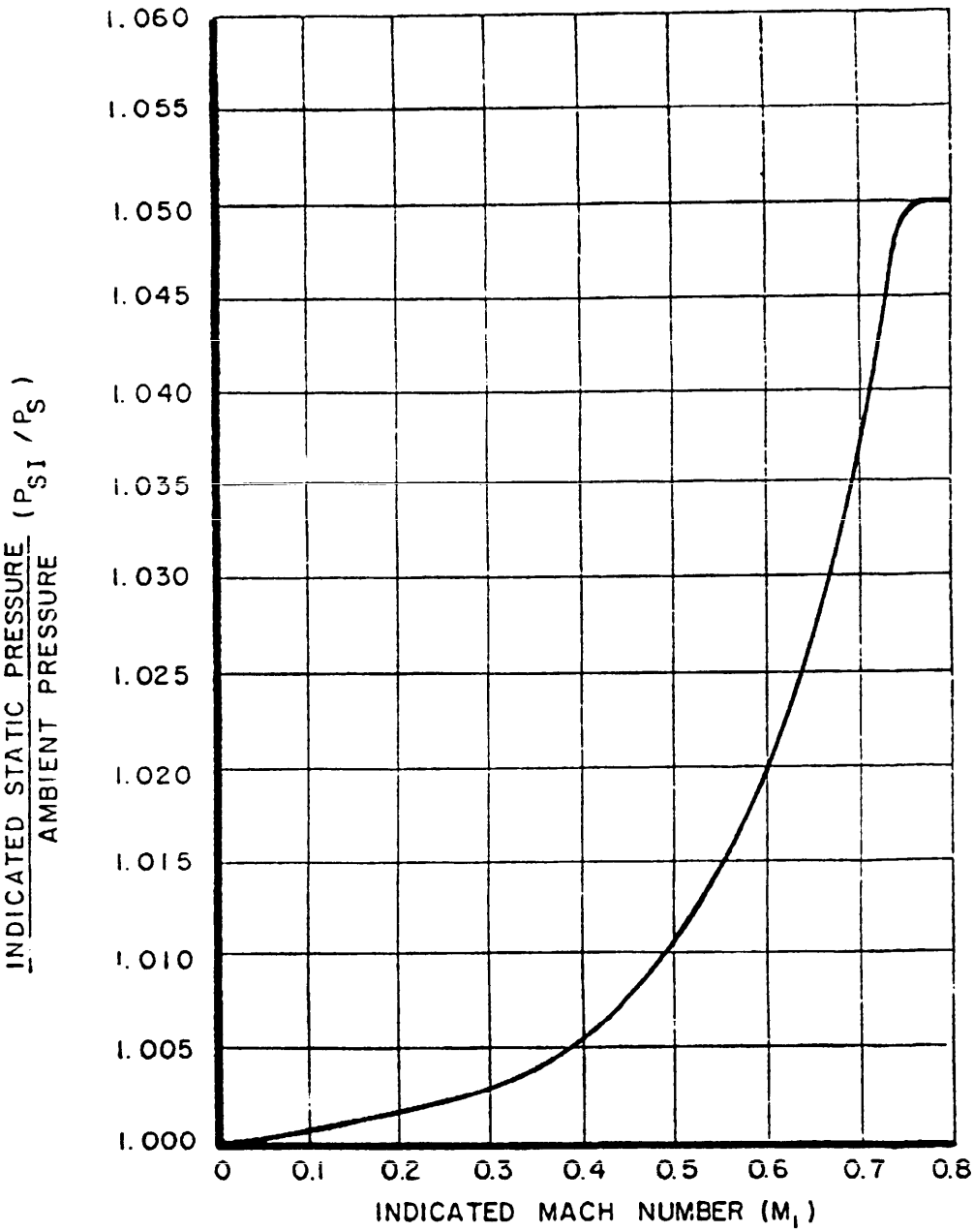


FIGURE 1. Pressure Error Correction

☆ U. S. GOVERNMENT PRINTING OFFICE: 1989-393-063/S-558

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE 1973—714-866/3983 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:09Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

ASD/ENYES
Wright-Patterson AFB, OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:09Z
Check the source to verify that this is the current version before use.